Buy Low, Sell Right

Predicting Fair Market Value with Machine Learning

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Code Demonstration

Jupyter Notebook Workflow

1. Cleaned and filtered raw Craigslist data
2. Split data into training and testing sets
3. Trained and compared multiple models

The Used Car Market is a Margin Game

$840B+ INDUSTRY

Highly competitive market driven by pricing accuracy.

THE CHALLENGE

Dealers must quickly identify undervalued vehicles.

OUR GOAL

Build a machine learning model to predict fair market value.

Pricing mistakes directly impact dealership profit margins.

Dataset Challenges

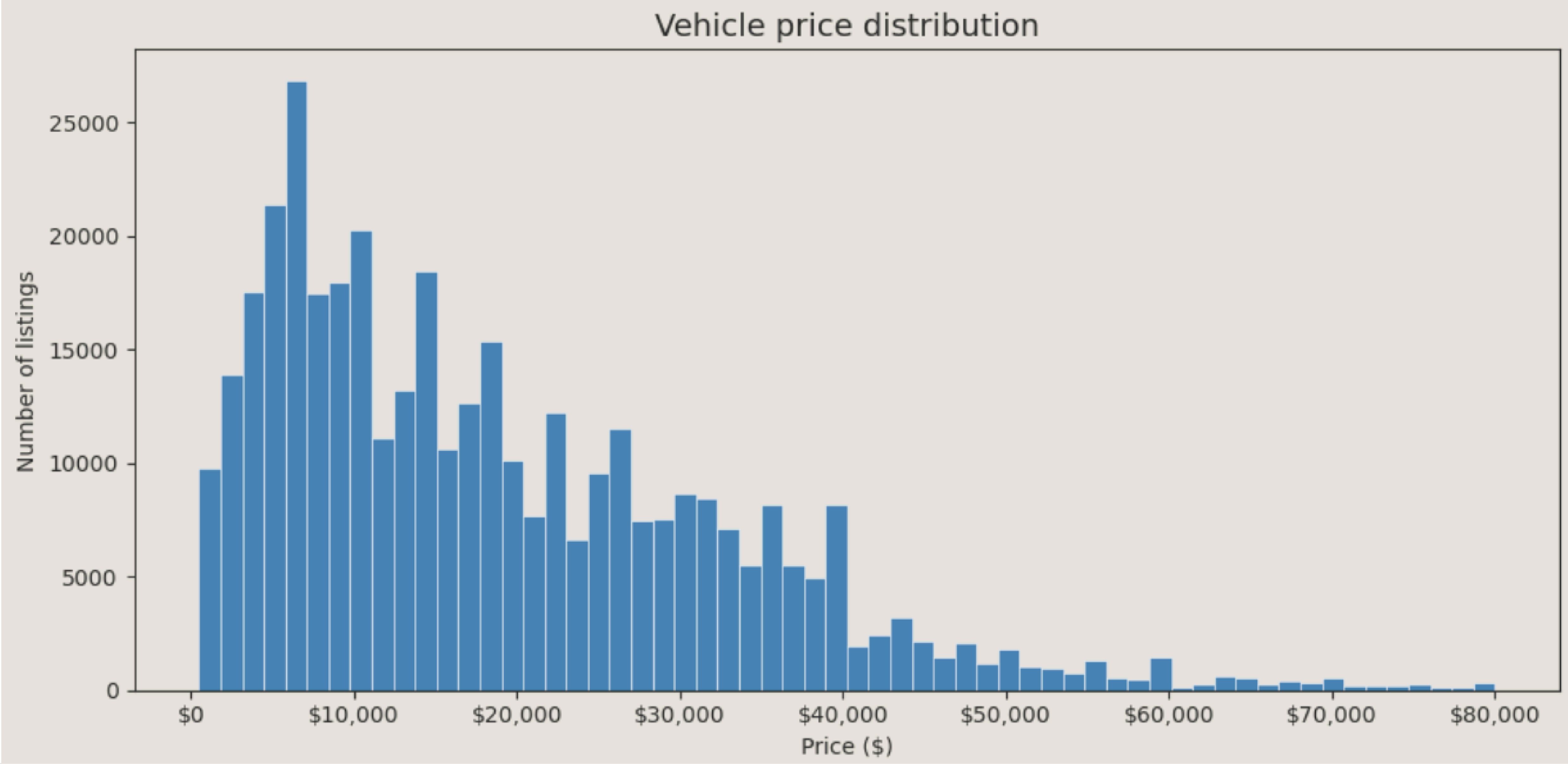
MAIN CHALLENGE

- Missing values
- Unrealistic prices
- Inconsistent listings

SECONDARY CHALLENGE

- Large data volums
- Outliers
- Model sensitivity

BEFORE CLEANING



Data Preparation

Removed unrealistic prices

↓

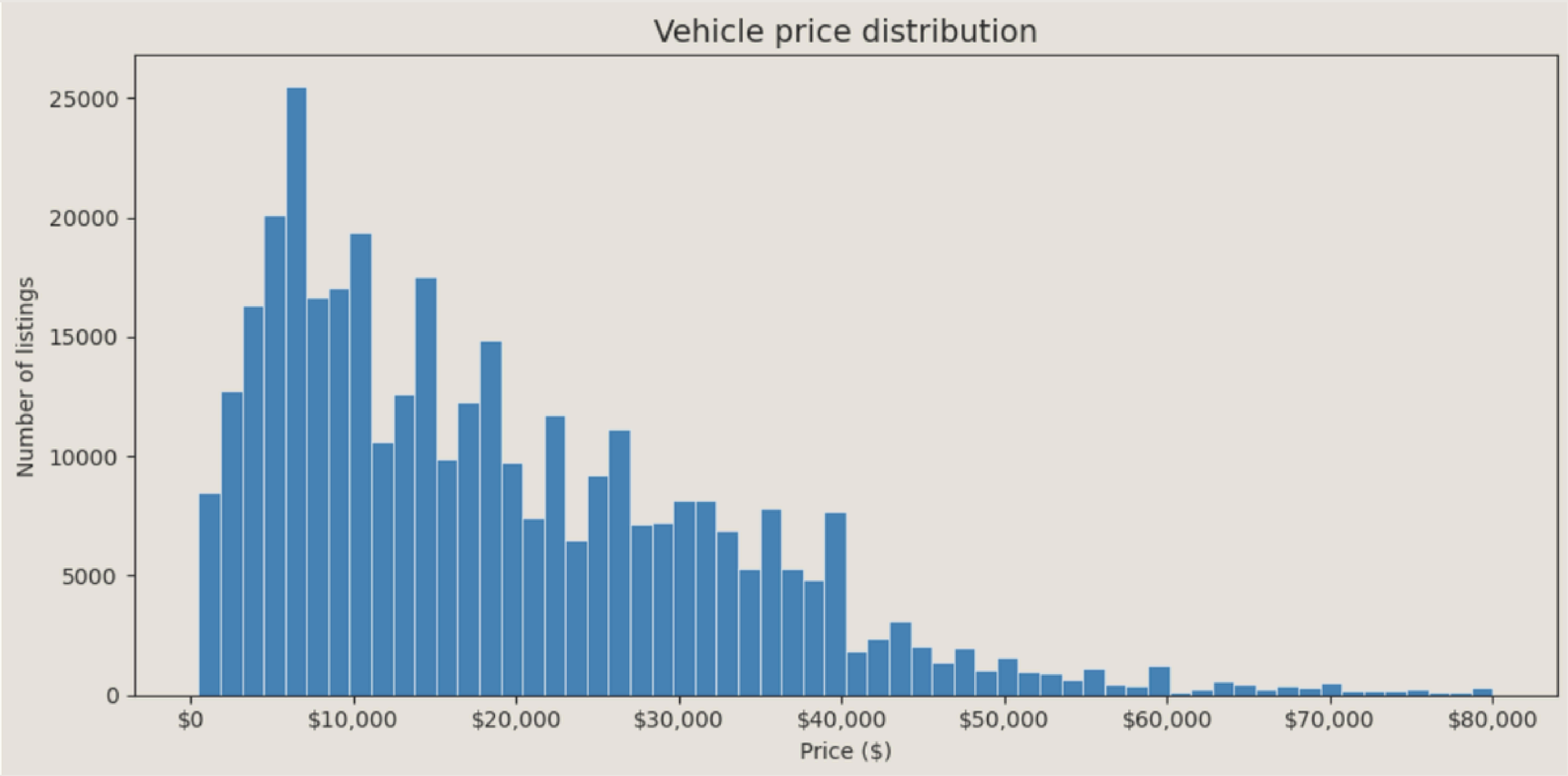
Removed incomplete columns

↓

Handled missing values

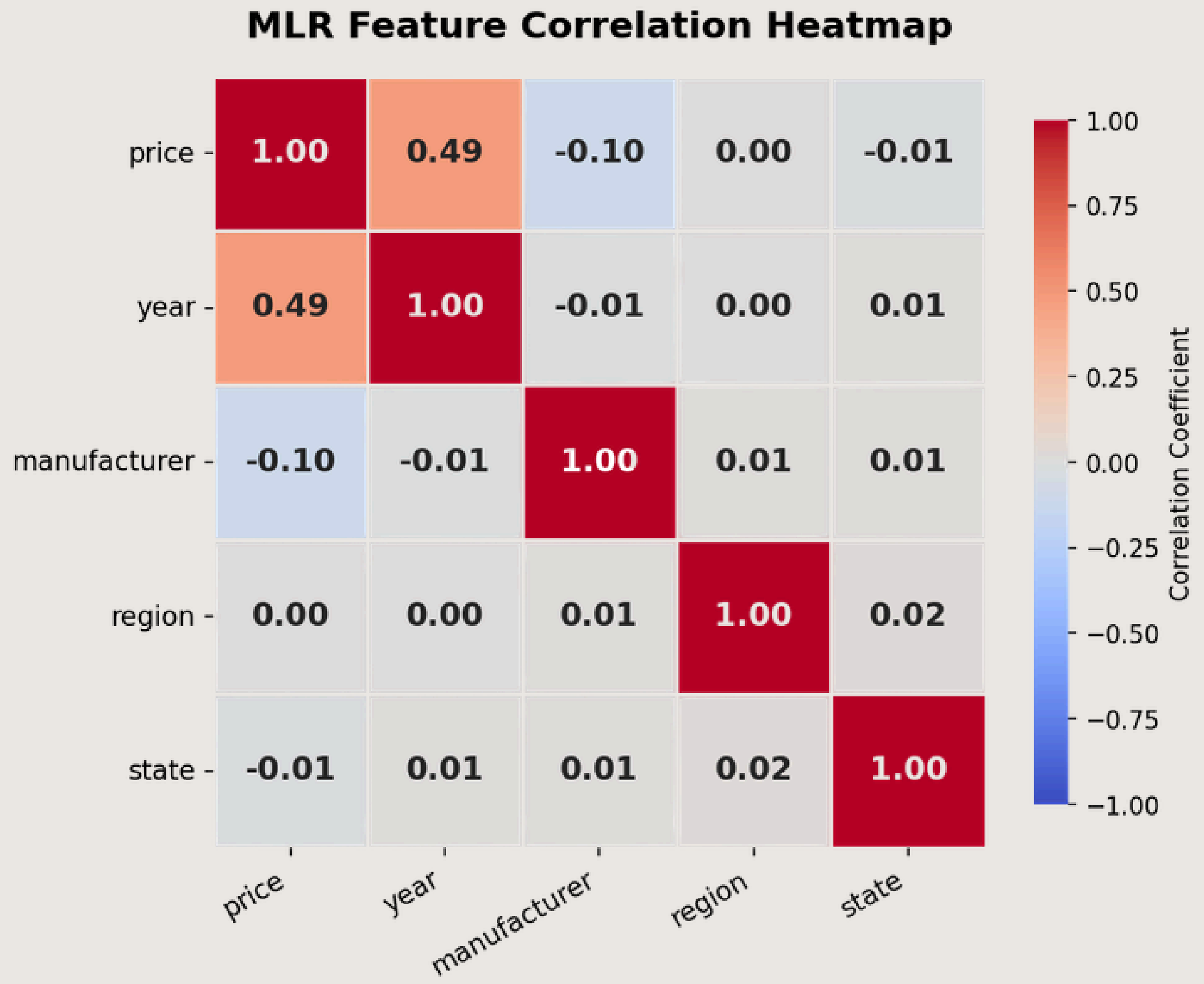
↓

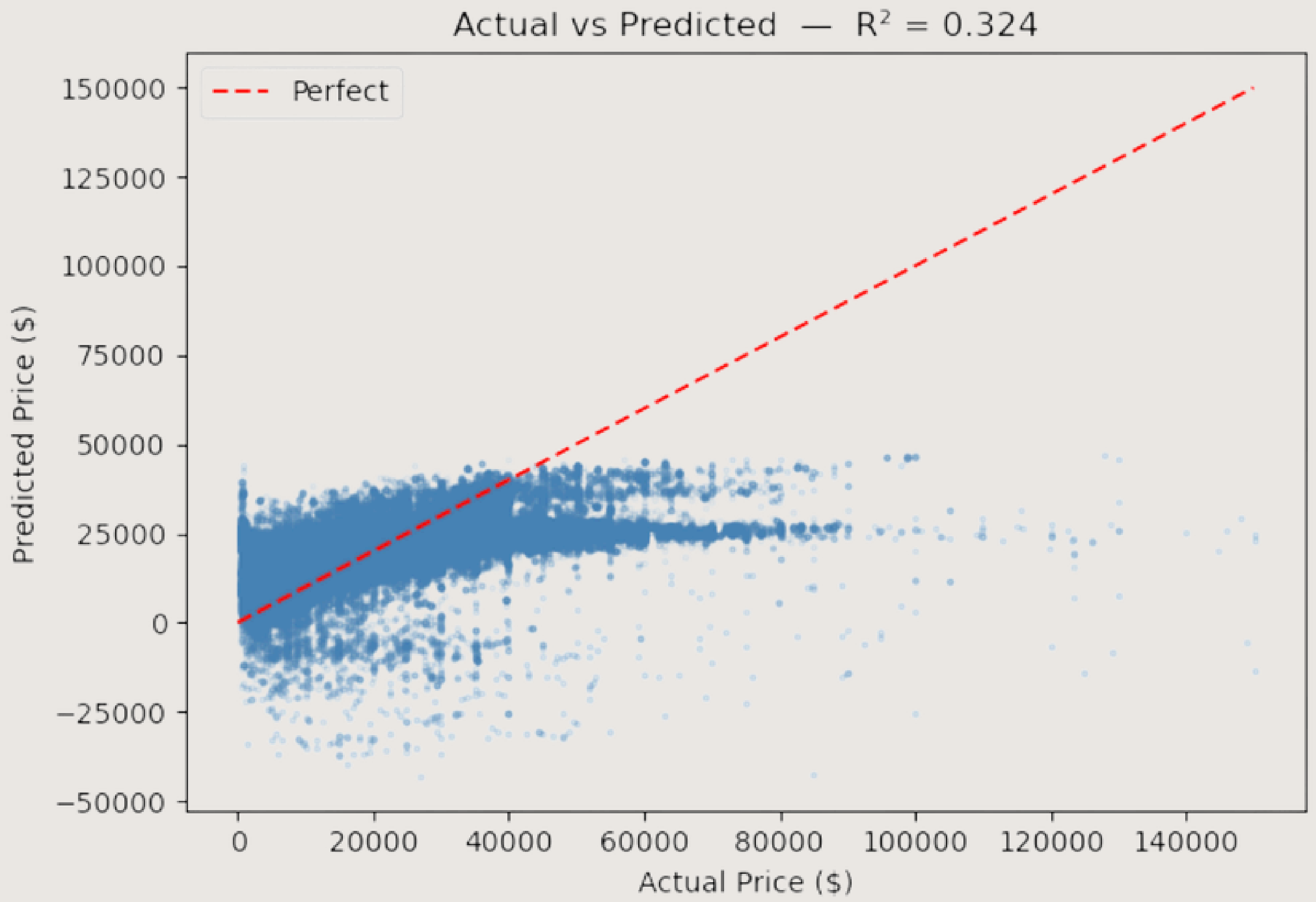
Prepared dataset for modeling



- 426,880 original listings
- **22,860 rows removed**
- unrealistic prices and missing values filtered out

Multiple Linear Regression





HOW IT WORKS

- Uses multiple variables to predict price
- Assumes linear relationships
- Easy to interpret

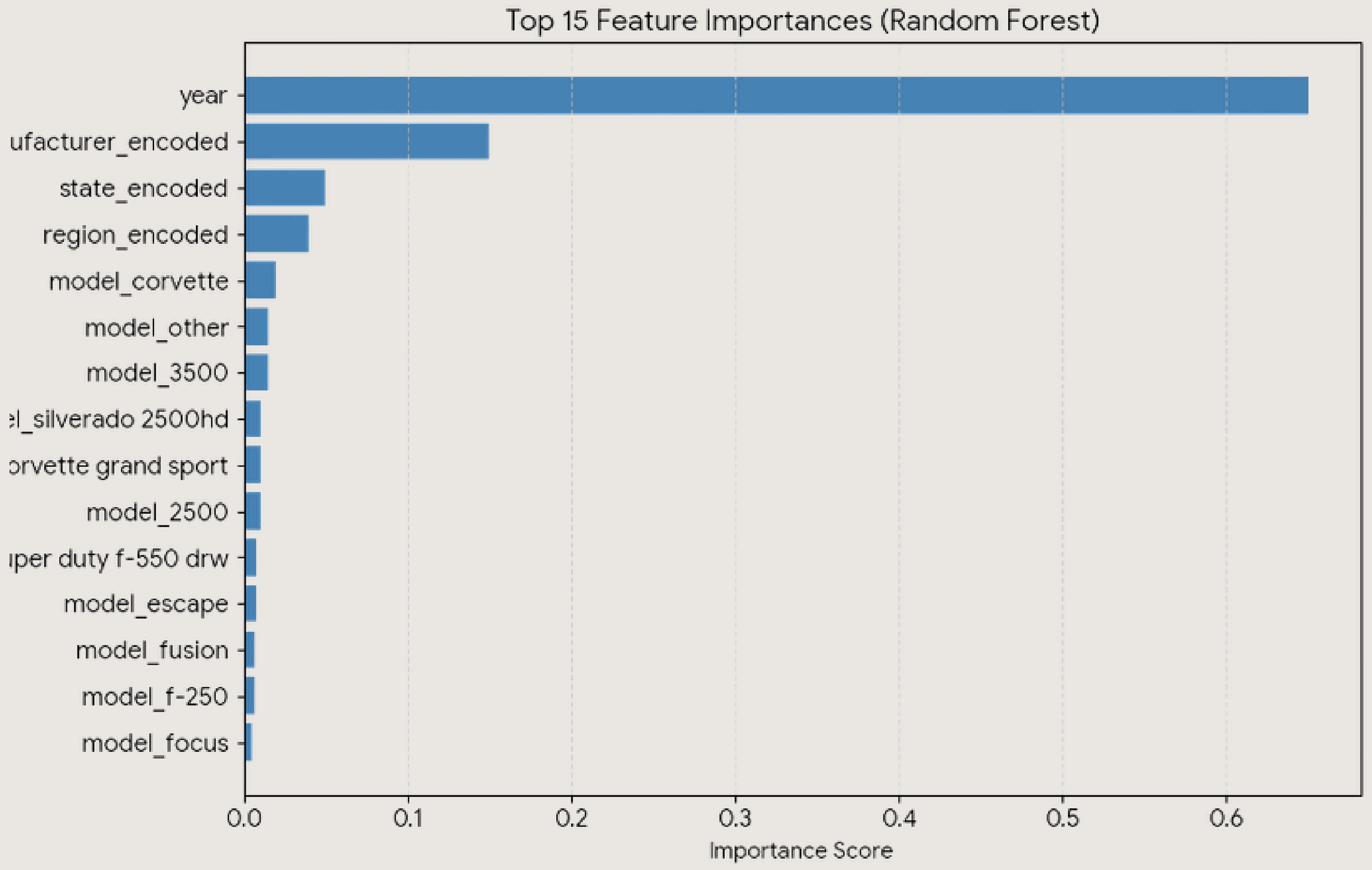
STRENGTHS

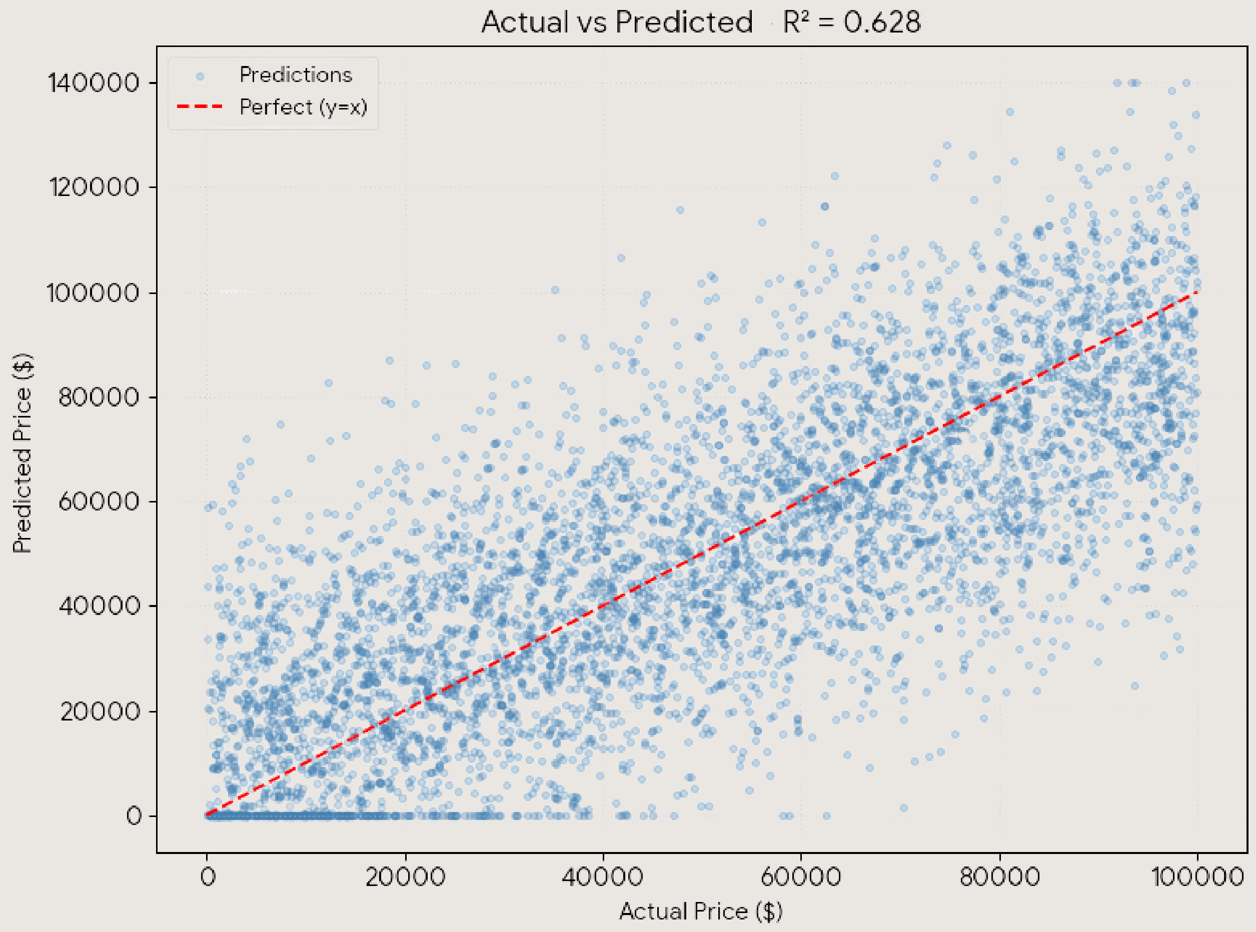
- Strong baseline model
- Fast and explainable
- Clear feature relationships

LIMITATION

- Less flexible with complex pricing
- Struggled with nonlinear patterns

Random Forest





HOW IT WORKS

- Combines multiple decision trees
- Averages predictions across models
- Reduces overfitting

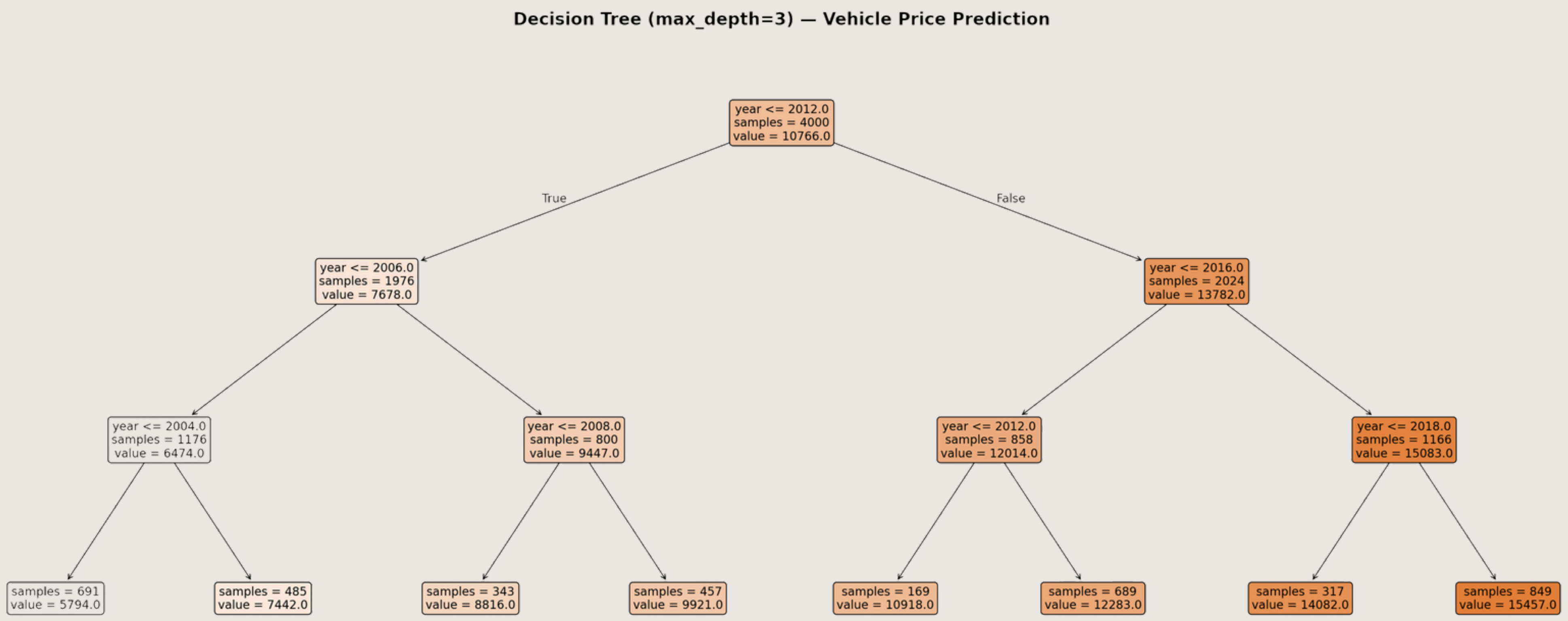
STRENGTHS

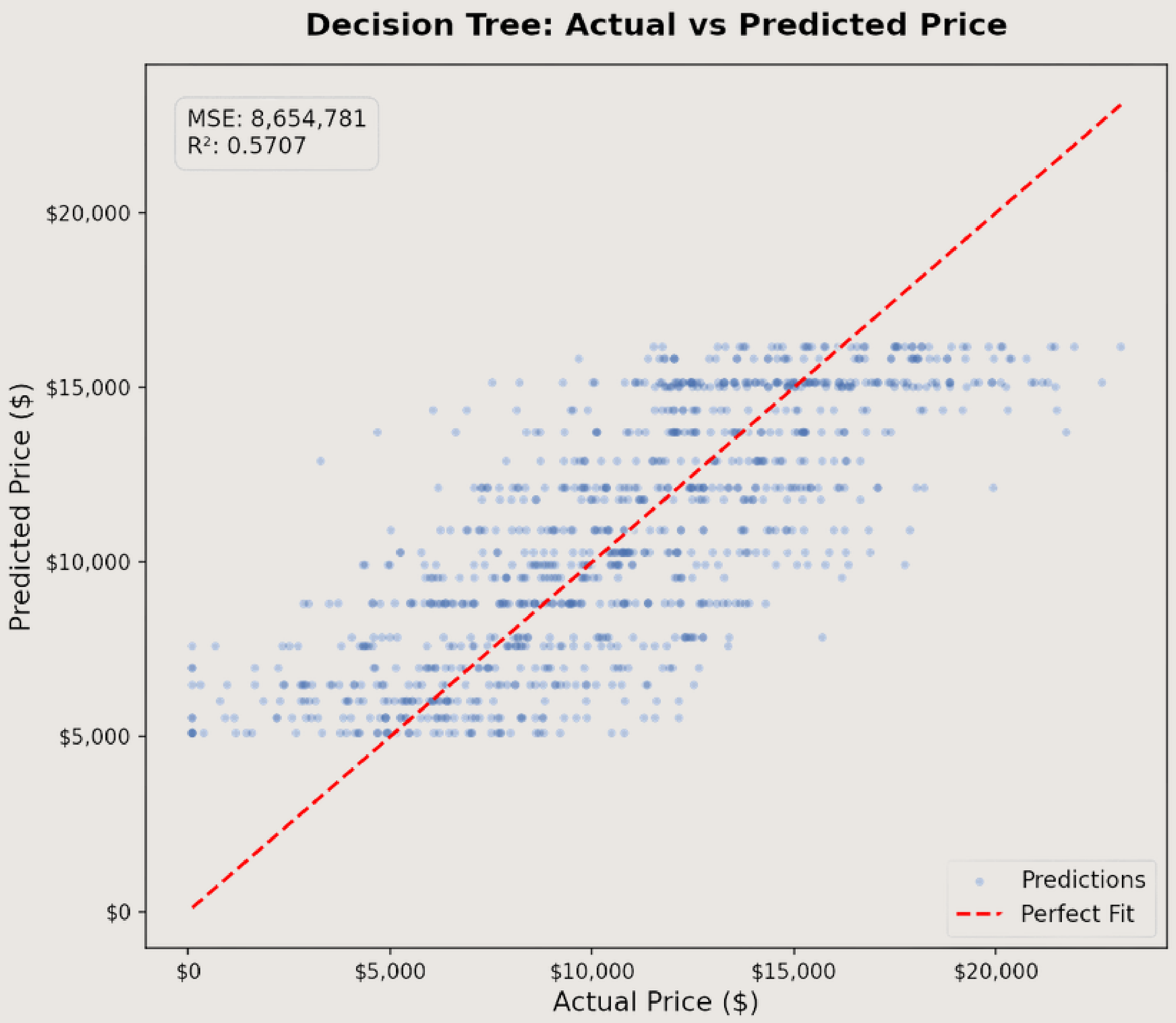
- Most accurate overall model
- Handles complex pricing patterns
- More stable predictions

LIMITATION

- More computationally complex
- Harder to interpret

Decision Tree





HOW IT WORKS

- Uses branching decisions to predict price
- Splits data into smaller groups
- Works similarly to a flowchart

STRENGTHS

- Captures nonlinear pricing behavior
- More flexible than regression
- Handles pricing complexity better

LIMITATION

- Risk of overfitting
- Can become too specific to training data

Which Model Performed Best?

Model	Strength	Weakness
Multiple Linear Regression	Simple & interpretable	Too rigid
Decision Tree	Captures complexity	Overfitting risk
Random Forest ⭐	**Highest accuracy**	Harder to interpret

- Multiple models were explored during analysis
- Random Forest performed best overall
- Lower mileage and newer vehicles increased value

Lessons Learned

WHAT WORKED

- Cleaning improved model performance
- Comparing models improved analysis
- Random Forest performed best overall

CHALLENGES

- Messy real-world data
- Missing values and outliers
- Balancing overfitting vs underfitting

Data preparation was just as important as the machine learning models themselves.

Thank you!